Neeraj Singh Guleria

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SUMMARY

In 2+ years as a Data Engineer at Optiemus Electronics, I designed and owned the full pipeline stack that now processes 1 million+ manufacturing records every month — from raw MES/ERP ingestion through ADF, PySpark transformations in Databricks, Delta Lake medallion layers, and Power BI delivery to senior leadership. That pipeline cut analysis time by 90% and recovered 100+ engineering hours a month. My stack is Azure-first — ADF, Databricks, Synapse, Delta Lake, dbt, Great Expectations, and Terraform — though the principles I work with translate directly across cloud platforms including AWS. I build pipelines that are tested before they ship, monitored after they do, and reproducible across environments. I take data quality and governance seriously enough to have built the guardrails proactively — schema validation, freshness checks, RBAC, and lineage tracking are all part of how I work, not optional extras. Currently targeting roles where infrastructure reliability and data quality are treated as first-class concerns, not afterthoughts.

WORK EXPERIENCE

Data Engineer — **Optiemus Electronics Ltd.**, Noida, India *March 2024 – Present*

- **Cut production analysis time by 90%** by replacing shift-level manual log reviews with an automated ADF → Databricks → Synapse pipeline that processes 1M+ records every month overnight — what took engineers 3–4 hours per shift now requires zero human effort.
- **Saved 100+ engineering hours every month** by automating yield, OEE, and defect reporting across SMT, Final Assembly, and Packaging lines — eliminating the need for per-line manual data pulls and standardising KPI reporting for production heads and senior management.
- **Architected the full Bronze → Silver → Gold medallion pipeline from scratch** on Azure Data Lake Storage Gen2 and Delta Lake — Bronze for raw landing, Silver for PySpark cleansing and enrichment, Gold for star-schema aggregations — serving defect tracking, yield analysis, and OEE monitoring across all manufacturing lines.
- **Reduced pipeline reprocessing overhead to near zero** by switching from full-load ADF runs to incremental pipelines using watermarking and tumbling window triggers — only net-new records are processed each cycle.
- **Eliminated untested SQL transformations** by introducing dbt across all Gold-layer logic — every model now has schema tests and source freshness checks; no transformation ships to production without passing them first.
- **Enforced data quality at the pipeline gate** using Great Expectations — null checks, schema assertions, and business-rule validations run before data is promoted to Silver or Gold, so broken data never reaches dashboards.
- **Made all infrastructure reproducible** by bringing ADF pipelines, Databricks clusters, Synapse pools, and Key Vault secrets under Terraform — dev, staging, and production environments are now identical and version-controlled.
- **Reduced pipeline failure detection from hours to 2 minutes** by building Grafana + Azure Monitor observability with automated alerting — on-call engineers receive failure context immediately, eliminating manual log hunting.
- **Designed a star-schema Synapse SQL dedicated pool** with Fact Production, Fact Defects, Dim Machine, and Dim Product tables — the data model backing all 4 live Power BI dashboards used daily by production heads and senior management.
- **Secured the full data layer** with Azure Key Vault for all credentials, RBAC policies for role-level access control, and Unity Catalog for data lineage tracking and metadata management across Databricks.
- **Halved MES extraction query time** by pushing complex filtering logic into PL/SQL stored procedures at the source database rather than pulling raw data and filtering in Spark — reducing ingestion overhead significantly.
- **Designed and scheduled pipeline** workflows using Databricks Jobs and Apache Airflow for automated ingestion and transformation.

Data Analyst Intern – **Living Things**, Mumbai, India *October 2023 – January 2024*

- **Automated weekly data validation workflows** using Python and SQL — standardised pipelines replaced manual Excel-based checks, reducing report preparation time for the operations team.
- **Surfaced operational insights from business datasets** that fed directly into leadership planning decisions — findings were communicated through Python visualisations to non-technical stakeholders.

Network Intern – Ekaga Group, Delhi, India *May 2022 – June 2022*

- Configured and tested a GPON network for an IPMPLS Defense Project, supporting IP address planning, hardware setup, and network troubleshooting.
- Gained hands-on exposure to network operations, TCP/IP protocols, and network security fundamentals.

SKILLS

Cloud Platforms: Microsoft Azure, Azure Data Factory (ADF), Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage Gen2 (ADLS Gen2), Azure Event Hubs, Azure Cosmos DB, Azure SQL Database, Azure Key Vault, Azure Purview, Snowflake

Big Data & Processing: Apache Spark, PySpark, Spark Structured Streaming, Apache Kafka, Delta Lake, Apache Airflow

ETL & Data Engineering: dbt (Data Build tool), ETL/ELT Pipeline Design, Medallion Architecture, Incremental Loading, Watermarking, SCD Type 2, Metadata-Driven Pipelines, Data Vault 2.0, REST API Integration

Databases & Warehousing: SQL (Advanced), PL/SQL, Azure Synapse Dedicated SQL Pools, Oracle, MySQL, Star Schema, Snowflake Schema

Programming: Python, PySpark, Pandas, NumPy, SQL, PowerShell, Bash, C#, JavaScript

Data Modelling & Governance: Star Schema, Snowflake Schema, Data Vault 2.0, RBAC, Data Lineage, Unity Catalog, Azure Purview

DevOps & CI/CD: GitHub Actions, Terraform, Azure DevOps, ARM Templates, Bicep, Docker

BI & Monitoring: Power BI, Azure Monitor, Log Analytics

Data Formats: Parquet, Avro, JSON, Delta Lake

PROJECTS

Azure Manufacturing Data Pipeline

- Problem: Manual production log analysis took several hours per cycle, creating delays in defect detection and yield reporting for 1 million+ monthly records.
- Solution: Built a complete ETL pipeline using Azure Data Factory, Azure Databricks (PySpark), and Azure Synapse Analytics with medallion architecture, Delta Lake optimization, data validation checks, and Azure Monitor alerting.
- Outcome: Reduced analysis time by 90%, enabled real-time Power BI dashboards, and fully automated reporting — recovering 100+ engineering hours per month.

MES Web Application for Self-Serve Production Reporting

- Problem: Production teams depended on IT to run complex SQL queries for historical data, creating bottlenecks and delays in operational decisions.
- Solution: Developed an ASP.NET Core MVC web application with dynamic filters, server-side pagination, and optimized PL/SQL stored procedures for fast data retrieval.
- Outcome: Reduced manual query turnaround time by 80%, enabling production teams to self-serve real-time reports without IT involvement.

End-to-End Azure Data Engineering Pipeline

- Problem: Organization lacked scalable data infrastructure for business intelligence reporting and executive KPI tracking.
- Solution: Designed and implemented full medallion architecture using ADF, Azure Databricks, Synapse Analytics, and Power BI, including incremental loads, SCD Type 2 handling, and star schema modelling.
- Outcome: Delivered 3 executive dashboards backed by 5 fact tables and 12 dimension tables, improving data accuracy, consistency, and reporting speed across teams.

EDUCATION

Bachelor of Science (Hons.) – Data Science *2020 – 2023*

K.R. Mangalam University, Gurgaon, Haryana

Relevant coursework: Statistical Data Analysis, Python, SQL, Data Mining, Machine Learning, Power BI, Database Management Systems (DBMS)

CERTIFICATIONS

- Azure Data Engineering Masters: Build Scalable Solutions – Udemy (2025)
- Azure Data Engineering for Data Engineers: DP-203 and Beyond – Udemy (2026)